Adaptive Anchor-Based Detector With Constrained RIRConv for Oriented Vehicles in SAR Images

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***Abstract*— In the field of synthetic aperture radar (SAR) vehicle detection, detectors based on deep learning have gained extensive application in recent years. However, most methods can only generate horizontal bounding boxes (HBBs). Due to the lack of angle information, the presence of excessive background noise in HBB leads to an inaccurate description of vehicles. Although research has been conducted on oriented SAR target detection, several issues persist, including the inadequate utilization of vehicle characteristics, inaccurate feature extraction for vehicles exhibiting size and rotation variations, and imprecise positioning of oriented bounding boxes (OBBs). To address these issues, we propose a novel-oriented SAR vehicle detection network. First, leveraging the shape property of vehicles, we devise a constrained rectangular-invariant rotatable convolution (Cons-RIRConv) for the backbone. Under the constraints of a carefully designed size- and rotation-invariant regularization term, Cons-RIRConv determines convolution sampling positions that are adaptive to the size and rotation variations of vehicles. This approach is conducive to Cons-RIRConv in extracting size- and rotation-invariant features of vehicles. Furthermore, as the sampling positions of Cons-RIRConv align closely with vehicles, we integrate them into the anchor generation process within the detection head, thereby developing an adaptive anchor generation mechanism (AAGM). Guided by the sampling positions of Cons-RIRConv, the anchor boxes generated by AAGM exhibit a high degree of conformity with the vehicles and low redundancy, ultimately improving the localization accuracy of OBB and reducing computation costs. Experiments on three authoritative measured SAR vehicle detection datasets show the effectiveness of our method.**

***Index Terms*— Adaptive anchor generation mechanism (AAGM), constrained rectangular-invariant rotatable convolution (Cons-RIRConv), synthetic aperture radar (SAR), vehicle detection.**

NOMENCLATURE

Δw, Δh	Scaling factors for convolutional sampling positions.
θ	Rotation angle for convolutional sampling positions.
K	Size length of the convolution kernel.
P, P'	Sets of original and transformed sampling positions.
$\hat{P}$	Target set of transformed sampling positions.
$\mathbf{c}_m$	Center position of the mth ground truth box.
Ab	Rotated anchor box.
L_{CLS}, L_{REG}	Classification loss and regression loss.
L_{SRI}	Size- and rotation-invariant regularization term.
L_s, L_{ch}	The smooth $L1$ loss and chamfer loss
$\mathbf{C}$, $\mathbf{P}$	Feature map tensor.
$\mathbf{R}$, $\mathbf{S}$	Rotation matrix and scaling matrix.

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I. INTRODUCTION

SYNTHETIC aperture radar (SAR) boasts numerous advantages, including its capability for continuous, all-weather observations. It has become a crucial piece of technical equipment being vigorously developed by countries worldwide. SAR target detection [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] is an important branch of SAR applications. In the early stages of research, scholars mainly focused on the detection of ships [3], [4], aircraft [5], [6], and other large-sized targets [7], [8]. Nowadays, with the steady enhancement of SAR image resolution, the detection of vehicle targets [9], [10] has become achievable. Consequently, this article focuses on vehicle detection in SAR images, a task that holds important significance in traffic management, route planning, and traffic flow detection.

A classic approach in SAR target detection is the constant false alarm rate (CFAR) method [1], [2], which utilizes clutter statistical modeling to detect targets by analyzing scattering intensity disparities between the target and its adjacent background. In addition, visual attention-based methods [10], [11] and wavelet transform-based methods [12] are also widely employed in SAR target detection. However, for vehicle targets, the scenes where they are located, e.g., urban and mountainous areas are very complex. Heterogeneous backgrounds significantly interfere with target detection. Under such conditions, the detection performance of these traditional methods is quite limited.

Currently, as computational capabilities continue to enhance and large-scale annotated datasets become increasingly accessible, deep learning has exhibited impressive performance in

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target detection, primarily attributed to its powerful feature extraction abilities. The detectors based on deep learning can be classified into horizontal bounding box (HBB)-based detectors (e.g., SSD [13], R-CNN series [14], [15], and Yolo series [16], [17]) and oriented bounding box (OBB)-based detectors (e.g., RRPN [18], CSL [19], and R^3Det [20]), depending on whether the detected bounding box incorporates the angle information of the target. Many researchers have refined these detectors and successfully applied them to SAR target detection [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32]. Among the HBB-based methods, Guo et al. [6] introduced a novel SAR aircraft detection network, which combined scattering information enhancement and attention mechanism to address the discreteness, variability, and interference of aircraft. Zhang et al. [21] proposed a balance learning method that effectively addresses issues pertaining to image scene imbalance, ship scale imbalance, and positive–negative sample imbalance in SAR ship detection. Du et al. [22] designed a semi-supervised network based on scene characteristic learning for SAR ship detection, significantly reducing the dependency of network training on target-level annotations. Compared with HBB, OBB offers distinct advantages, including reduced interference from background pixels and a more precise estimation of target length and width [26]. In addition, in scenes where targets are densely arranged, HBB is more likely to overlap with each other, resulting in a higher rate of missed detections following nonmaximum suppression compared to OBB-based methods. Consequently, SAR targets with arbitrary angles and dense arrangements are more aptly represented by OBB. Researchers have proposed many OBB-based SAR target detection methods. Notably, the DRBox series [26], [27] were early proposed oriented SAR target detectors and have demonstrated impressive performance. Fu et al. [28] designed an oriented SAR ship detection network that integrates SAR imaging principles, enabling it to achieve excellent performance across diverse imaging conditions. Gao et al. [29] exploited the scattering and geometric characteristics of targets to design a novel OBB-based SAR ship detection network, effectively addressing the issue of boundary discontinuity in the prediction of target angles. Zhang et al. [30] developed a hierarchical detection strategy for multiscale ships in the feature pyramid network, thereby enhancing the robustness of the model in detecting ships of different sizes. Due to the weak texture features of SAR bridges, Chen et al. [32] utilized dynamic global attention to enhance the features of bridges, achieving more accurate localization of SAR bridges. These aforementioned methods have played an important role in propelling the progress of SAR target detection.

However, there remain several issues that necessitate addressing in OBB-based SAR target detection methods. First, most OBB-based SAR target detection methods primarily concentrate on targets such as ships, aircraft, and buildings. Compared to stationary targets, such as buildings, vehicles typically exhibit smaller physical sizes. The features of vehicles are more susceptible to the influences of the surrounding environment, leading to less distinct target contours and texture features. Compared to other time-sensitive targets, such as ships and aircraft, vehicles are typically densely arranged in complex scenes, including parking lots and urban highways, where problems of multitarget interference and clutter interference are more prominent. If the detection methods designed for ships, aircraft, and buildings are directly applied to the SAR vehicle detection, it is difficult to obtain satisfactory results. Therefore, how to leverage the characteristics of vehicles, such as the rectangular shape, size range, and other properties, to design targeted SAR vehicle detection methods is worthy of further study.

Second, vehicle targets within SAR images exhibit significant variations in size and orientation, and are often densely arranged. Taking the vehicles from the two datasets, miniSAR [33] and FARADSAR [34], as examples, Fig. 1 presents the distribution histograms of pixel length, pixel width, pixel area, and rotation angle (defined by the longest side method [35]) for the bounding boxes of vehicles in the combined training data derived from the two datasets. It is evident that there are size variations in the pixel length and pixel width of different vehicles, particularly in pixel length, where the difference between the maximum and minimum pixel lengths of vehicles exceeds 70. The pixel areas of different vehicles also exhibit significant differences, with the disparity between the largest and smallest pixel areas exceeding 3000. As observed in Fig. 1(d), vehicles are distributed across various angle ranges, and the angular differences among different vehicles are also quite pronounced. The convolution operation in the traditional convolutional layer is limited by its fixed-size and square-shaped sampling positions. This limitation can result in interference from surrounding clutter and other vehicles when the traditional convolutional layer extracts features from a specific vehicle. Therefore, the extraction of size- and rotation-invariant features of vehicles is of great importance. Performing data augmentation on training data, such as multiangle rotation and multiscale scaling, can enhance the model's adaptability to vehicles of varying orientations and sizes. However, since the convolutional sampling mechanism remains unaltered, the degree of performance improvement is constrained. Certain specialized operations [36], [37], which have been enhanced based on the convolution operation, aim to address this issue through the modeling of geometric variations in the target. For instance, deformable convolution [36] adapts to the geometric variations of the target by incorporating learnable 3-D offsets into each sampling position. However, due to the lack of clear supervision signals, their modeling results are not very satisfactory.

Third, compared with HBB, OBB needs to additionally predict the orientation angle of the target. Even minor errors in angular regression can potentially lead to inaccuracies in OBB positioning, subsequently affecting detection performance. Therefore, OBB poses a greater regression challenge and places higher demands on localization accuracy. The majority of anchor-based methods [18], [19], [20] strive to mitigate the complexity of OBB regression by densely sampling a plethora of anchor boxes across the angular range of 0°–180°. However, this gives rise to significant redundancy in anchor boxes, exacerbating the imbalance between positive (targets) and negative

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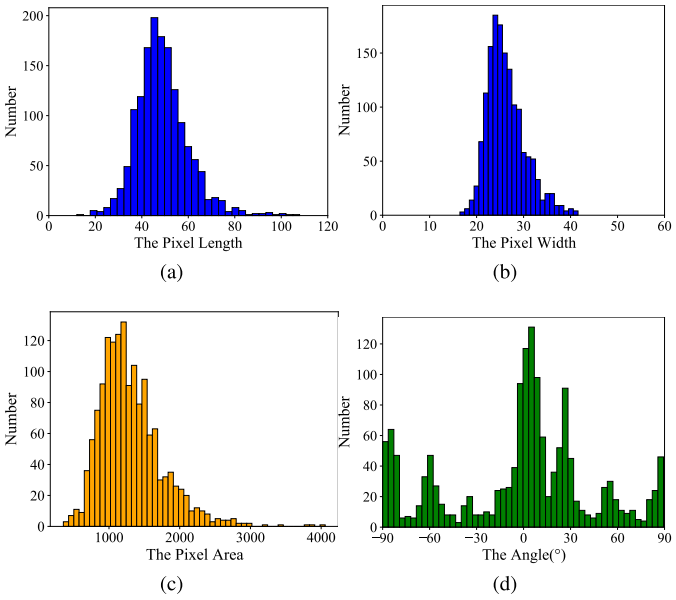

Fig. 1. Distribution histograms of pixel length, pixel width, pixel area, and rotation angle for the vehicles. (a) Distribution histograms of pixel length for the vehicles. (b) Distribution histograms of pixel width for the vehicles. (c) Distribution histograms of pixel area for the vehicles. (d) Distribution histograms of rotation angle for the vehicles.

(clutter) samples, and leading to substantial computational overhead.

To address the aforementioned issues, we propose a novel-oriented SAR vehicle detection network in this article. As depicted in Fig. 2, the entire network comprises four modules: input, backbone, feature pyramid, and detection head. Initially, the backbone serves to extract the deep multiscale features $\mathbf{C}_{r\in[1,2,3]}$ of the input SAR image. Notably, within this module, we have designed constrained rectangular-invariant rotatable convolution (Cons-RIRConv), which incorporates the shape property of vehicles. Specifically, the shapes of various vehicles in SAR images closely resemble rectangles but exhibit variations in size and orientation. Taking this into account, we configure three learnable parameters for Cons-RIRConv, i.e., Δw, Δh, and θ, for modeling the size and orientation variations of vehicles. These parameters enable the sampling positions of Cons-RIRConv to undergo scaling and rotation transformations while maintaining a rectangular shape. Furthermore, the transformed sampling positions are constrained by an explicit supervisory signal, specifically, a well-designed size- and rotation-invariant regularization term, to ensure a high degree of conformity with the vehicles. By doing so, the size- and rotation-invariant features of vehicles can be accurately extracted. Then, in the feature pyramid, the multiscale features $\mathbf{C}_{r\in[1,2,3]}$ are integrated to generate $\mathbf{P}_{r\in[1,2,3]}$ with richer contextual information, which serves as the input for the detection head. For the value range of r, we take into full account the size property of vehicles. Finally, we integrate the convolution sampling positions with the anchor generation process in the detection head, thereby designing an adaptive anchor generation mechanism (AAGM) for bounding box regression and classification. This mechanism eliminates the need for manually designed dense anchors. Instead, guided by the convolution sampling positions of Cons-RIRConv, the AAGM is able to adaptively generate a smaller number of anchor boxes that closely match vehicles. This configuration of anchor boxes effectively alleviates the difficulty of regression tasks, consequently improving the localization accuracy of OBB. Moreover, the reduction in the number of anchor boxes significantly mitigates anchor redundancy, thereby improving computing efficiency and mitigating the imbalance between positive and negative samples.

The main contributions of this article are as follows.

1) We propose a well-designed oriented SAR vehicle detection network in this article. During the design process, the geometric properties of vehicles, i.e., the shape and size properties are fully utilized. Compared to other advanced OBB-based methods, our proposed approach demonstrates superior detection performance.
2) We propose Cons-RIRConv, which is capable of extracting size- and rotation-invariant features of vehicles by adaptively adjusting the convolution sampling positions. Cons-RIRConv exhibits greater adaptability to vehicles with variations in size and orientation, thereby enhancing the feature representation capabilities of the backbone for vehicles.
3) By incorporating the convolution sampling positions of Cons-RIRConv, we present an AAGM. This mechanism eliminates the need for manually designed dense anchors. The anchor boxes generated by AAGM demonstrate high conformity with the vehicles and reduced redundancy. Consequently, AAGM contributes to improving the localization accuracy of OBB and accelerating network convergence.

We conducted extensive experiments utilizing three measured SAR vehicle detection datasets: the SIVED dataset [35], the miniSAR dataset [33], and the FARADSAR dataset [34]. These experiments were designed to show the effectiveness of our method. The organization of the subsequent sections of this article is outlined as follows. In Section II, we elaborate on the proposed method. In Section III, we discuss the experimental results on the three datasets. Finally, Section IV summarizes the conclusion.

II. Methodology

A. Overall Framework

The primary mathematical symbols utilized in this article are summarized in Nomenclature. In this section, we will introduce the backbone, feature pyramid, and detection head of the network.

1) Backbone: Positioned after the input SAR image, the backbone serves as the fundamental component of the entire network. Its primary function is to extract deep features from the SAR image. Notably, we have incorporated the Cons-RIRConv into the backbone to extract size- and rotation-invariant features of vehicles. The design specifics of the Cons-RIRConv will be delved into in Section II-B.

2) Feature Pyramid: The feature pyramid utilizes the outputs $\mathbf{C}_{r\in[1,2,3]}$ of the three Cons-RIRConv stages in the backbone as inputs. Similar to [38], by combining high-level semantic features with low-level high-resolution features, the resulting fused feature $\mathbf{P}_{r\in[1,2,3]}$ can incorporate more comprehensive contextual information. This is beneficial for enhancing the capability of the model to detect multiscale

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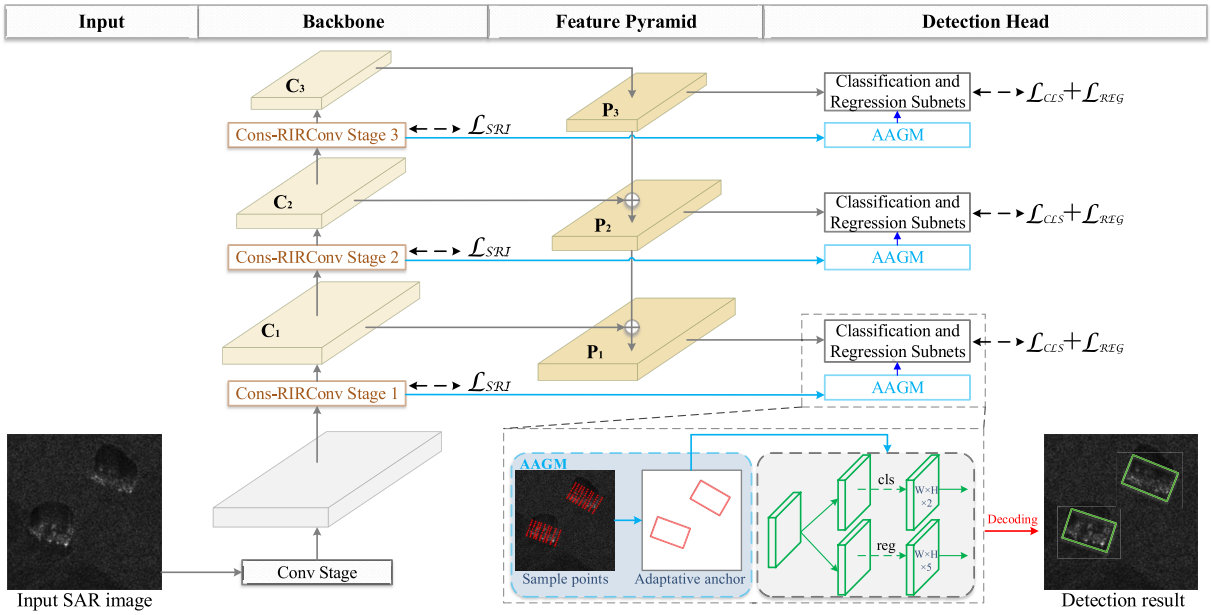

Fig. 2. Network structure of the proposed method.

vehicles. Please note that we refer to [39] to determine the range of values for r (i.e., $r \in [1, 2, 3]$) and the down-sampling factors s_r of $\mathbf{P}_{r\in[1,2,3]}$ relative to the original image (i.e., 1/8, 1/16, 1/32). More specifically, the value of s_r is related to the ideal target scale (ITS) of the feature maps and the target scale d_m^B

$$\mathrm{ITS}_r = K \times \frac{1}{s_r} \tag{1}$$

$$d_m^B = \sqrt{h_m^B \times w_m^B} \tag{2}$$

where K represents the size length of the convolution kernel, and h_m^B and w_m^B represent the height and width of the oriented ground truth box of target m, respectively. They are related to the size properties of vehicles in the real world (typically, the lengths range from 2 to 9 m, and the widths range from 1.6 to 3.5 m) and the SAR image resolution. To ensure that the ITS of the feature maps covers the distribution range of d_m^B as precisely as possible, we set r to 3, s_1 to 1/8, s_2 to 1/16, and s_3 to 1/32, based on (1) and (2). More detailed information can also be traced in [39].

3) Detection Head: The detection head comprises an AAGM, a classification subnetwork, and a regression subnetwork. First, the AAGM adaptively generates rotated anchor boxes based on the sampling positions of the Cons-RIRConv. Subsequently, the classification subnetwork performs binary classification (vehicle or clutter) for the rotated anchor boxes, while the regression subnetwork predicts the offset vectors between the rotated anchor boxes and the ground-truth boxes. During training, the detection loss comprises both classification loss L_{CLS} and regression loss L_{REG}

$$L_{\mathrm{CLS}} = -\frac{1}{N}\sum_{n=1}^{N} d'_n \times \log(d_n) + \left(1 - d'_n\right) \times \log(1 - d_n). \tag{3}$$

The L_{CLS} is the cross-entropy loss. N represents the number of rotated anchor boxes. d'_n represents the predicted probability. d_n represents the ground truth

$$L_{\mathrm{REG}} = \frac{1}{N}\sum_{n=1}^{N} t_n \frac{L_s\left(v'_n, v_n\right)}{\left|L_s\left(v'_n, v_n\right)\right|} \left| - \log(\mathrm{SkewIoU}) \right|. \tag{4}$$

The L_{REG} is the derivable approximate SkewIoU loss [20]. In (4), t_n is a binary value ($t_n = 1$ for vehicle and $t_n = 0$ for clutter). v'_n represents the predicted offset vector. v_n represents the target vector of ground truth. SkewIoU represents the overlap of the predicted box and ground truth box. L_s represents the smooth-$L1$ loss function.

In Sections II-B and II-C, we will present a comprehensive exposition of the Cons-RIRConv within the backbone and the AAGM within the detection head.

B. Constrained Rectangular-Invariant Rotatable Convolution

Vehicle targets in SAR images often exhibit variations in size and orientation. These variations can lead to interference from surrounding clutter and other vehicles when the traditional convolutional layer extracts features from a specific vehicle. As depicted in Fig. 3, it can be observed that when the convolutional layer extracts features from vehicles A and B, some of its sampling positions overlap with areas occupied by surrounding clutter and other vehicles. Consequently, this results in inaccurate feature extraction for both vehicles A and B. When utilizing larger convolutional kernels or in deeper layers of the network, this issue becomes particularly pronounced due to the enlarged receptive field of the feature maps. To mitigate this issue, we propose Cons-RIRConv by incorporating the shape property of vehicles. Constrained by a well-designed size- and rotation-invariant regularization term, Cons-RIRConv can adaptively transform its sampling positions to accommodate variations in size and orientation among different vehicles. The following provides a detailed

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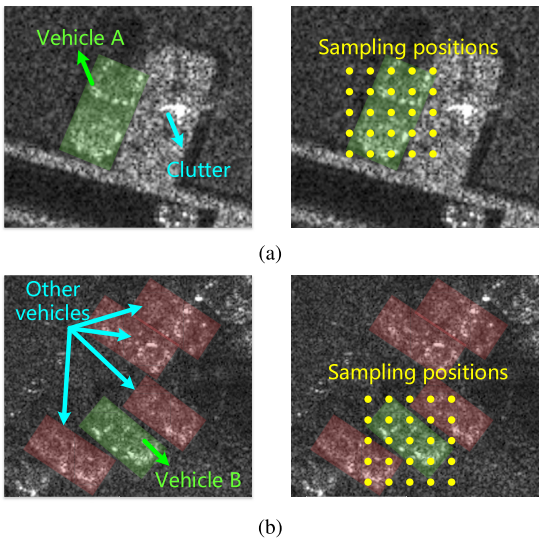

Fig. 3. Illustration that the convolutional layer cannot accurately extract vehicle features. (a) Sampling position overlaps with the surrounding clutter. (b) Sampling position overlaps with other vehicles.

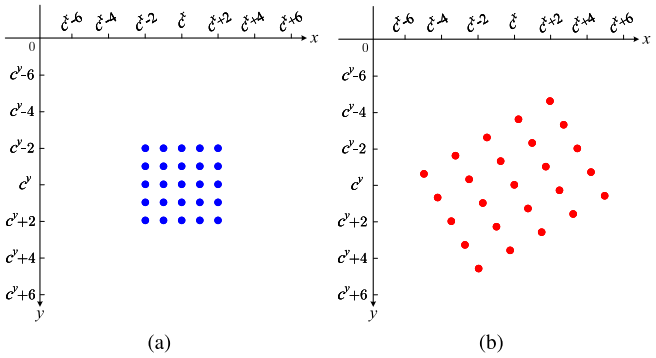

Fig. 4. Sampling positions of traditional convolution and Cons-RIRConv. (a) Sampling positions of traditional convolution. (b) Sampling positions of Cons-RIRConv when $\Delta w = 2$, $\Delta h = 1.5$, and $\theta = 30°$.

explanation of the transformation of sampling positions, along with the size- and rotation-invariant regularization terms.

1) Transformation of Sampling Positions: During the execution of convolutional operations, traditional convolutional layers display a fixed square-shaped sampling pattern. Assuming that the center position of sampling is designated as $\mathbf{c} = (c^x, c^y)$, the set of sampling positions can be represented by (5) and visualized in Fig. 4(a)

$$\begin{aligned} P_c &= \{\mathbf{p}_{ab} \mid a = 1, \ldots, K, b = 1, \ldots, K\} \\ &= \{\mathbf{g}_{ab} + \mathbf{c} \mid a = 1, \ldots, K, b = 1, \ldots, K\} \end{aligned} \tag{5}$$

where P_c represents the set of sampling positions when the center position is $\mathbf{c}$. K represents the side length of the convolution kernel. $\mathbf{p}_{ab}$ denotes the sampling positions, and $\mathbf{g}_{ab}$ denotes the offset of $\mathbf{p}_{ab}$ relative to the center position $\mathbf{c}$. Taking a 5×5 convolution kernel with dilation 1 as an example, the values of $\mathbf{g}_{ab}$ are as follows:

$$\begin{aligned} &\mathbf{g}_{11} = (-2, -2); \quad \mathbf{g}_{12} = (-1, -2); \quad \cdots \quad \mathbf{g}_{15} = (2, -2) \\ &\mathbf{g}_{21} = (-2, -1); \quad \mathbf{g}_{22} = (-1, -1); \quad \cdots \quad \mathbf{g}_{25} = (2, -1) \\ &\cdots \qquad\qquad \cdots \qquad\qquad \cdots \quad \cdots \end{aligned}$$

$$\mathbf{g}_{51} = (-2, 2); \quad \mathbf{g}_{52} = (-1, 2); \quad \cdots \quad \mathbf{g}_{55} = (2, 2). \tag{6}$$

In SAR images, various vehicles exhibit size and orientation variations, yet they consistently retain a rectangular shape. Considering this characteristic of vehicles, we reformulate the sampling positions. Specifically, in Cons-RIRConv, we introduce two transformation matrices, i.e., the rotation matrix $\mathbf{R}$ and the scaling matrix $\mathbf{S}$, to perform rotation and scaling transformations on the original sampling positions. Consequently, the set of transformed sampling positions is derived as follows:

$$\begin{aligned} P_c' &= \{\mathbf{p}'_{ab} \mid a = 1, \ldots, K, b = 1, \ldots, K\} \\ &= \{\mathbf{g}'_{ab} + \mathbf{c} \mid a = 1, \ldots, K, b = 1, \ldots, K\} \end{aligned} \tag{7}$$

$$\mathbf{g}'_{ab} = \mathbf{R}\mathbf{S}\mathbf{g}_{ab} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \Delta w & 0 \\ 0 & \Delta h \end{bmatrix} \mathbf{g}_{ab} \tag{8}$$

where Δw and Δh denote the scaling factors for the width and length of the convolutional kernel, respectively. θ represents the rotation angle of the sampling positions. Fig. 4(b) illustrates the sampling positions of Cons-RIRConv when $\Delta w = 2$, $\Delta h = 1.5$, and $\theta = 30°$. As can be observed from Fig. 4, with the inclusion of two transformation matrices, the sampling positions acquire the ability to rotate and scale, no longer adhering to the original fixed square sampling shape.

By summing the sampled values weighted by $\mathbf{w}$, the corresponding output feature value can be obtained. Taking the position $\mathbf{c}$ on the output feature map $\mathbf{C}_{\text{out}}$ as an example, we have

$$\begin{aligned} \mathbf{C}_{\text{out}}(\mathbf{c}) &= \sum_{1 \le a,b \le K} \mathbf{w}(\mathbf{g}_{ab}) \cdot \mathbf{C}_{in}\left(\mathbf{p}'_{ab}\right) \\ &= \sum_{1 \le a,b \le K} \mathbf{w}(\mathbf{g}_{ab}) \cdot \mathbf{C}_{in}\left(\mathbf{g}'_{ab} + \mathbf{c}\right) \end{aligned} \tag{9}$$

where $\mathbf{C}_{\text{in}}$ denotes the input feature map. As the transformed sampling positions $\mathbf{g}'_{ab}$ may be fractional. The feature values at fractional positions will undergo bilinear interpolation, relying on the feature values at integer positions.

2) Size- and Rotation-Invariant Regularization Term: After the transformation, the sampling positions acquire the ability to rotate and scale. In order to maximize the alignment between the transformed sampling positions and the vehicle, we devise a size- and rotation-invariant regularization term to optimize the transformed sampling positions. Notably, in devising the regularization term, we once again capitalize on the rectangular shape property of vehicles. Specifically, we observe that, due to the regular rectangular shape of vehicles, the pixels within the oriented ground truth box of the vehicle consist almost entirely of vehicle pixels, with minimal clutter pixels. Exploiting this characteristic, we generate a target set $\hat{P}$ from the oriented ground truth box for the set of transformed sampling positions. The size- and rotation-invariant regularization term is designed based on this target set. The target set $\hat{P}$ is represented as the following equation:

$$\hat{P} = \left\{\hat{\mathbf{p}}_{ij} \mid i = 1, \ldots, K, j = 1, \ldots, K\right\} \tag{10}$$

where $\hat{\mathbf{p}}_{ij} = (\hat{p}^x_{ij}, \hat{p}^y_{ij})$ denotes the positions in the target set $\hat{P}$. The values of $\hat{p}^x_{ij}$ and $\hat{p}^y_{ij}$ satisfy the following parameter

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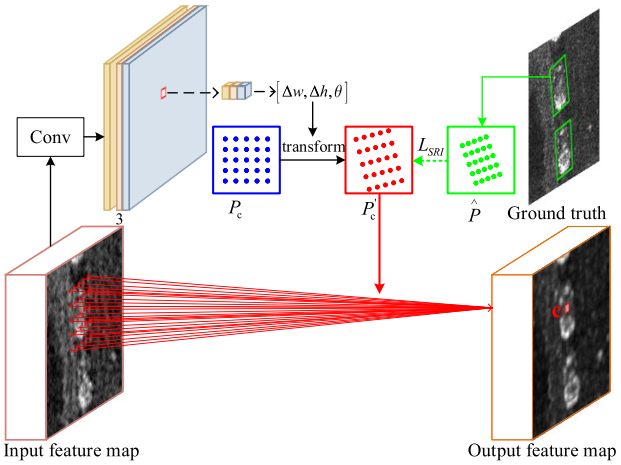

Fig. 5. Illustration of the 5 × 5 Cons-RIRConv.

equations:

$$\begin{cases} \hat{p}_{ij}^{x} = s_r \times \left(x_1 + \dfrac{i-1}{K-1}(x_4 - x_1) + \dfrac{j-1}{K-1}(x_2 - x_1) \right) \\ \hat{p}_{ij}^{y} = s_r \times \left(y_1 + \dfrac{i-1}{K-1}(y_4 - y_1) + \dfrac{j-1}{K-1}(y_2 - y_1) \right) \end{cases} \tag{11}$$

where (x_1, y_1), (x_2, y_2), (x_3, y_3), and (x_4, y_4) represent the positional coordinates of the upper-left, upper-right, lower-right, and lower-left corners of the oriented ground truth box, respectively. s_r denotes the down-sampling factors of the input feature map for Cons-RIRConv, relative to the input SAR image.

The size- and rotation-invariant regularization terms can be formulated as

$$\begin{aligned} L_{\text{SRI}} &= \frac{1}{M}\sum_{m=1}^{M} L_{\text{ch}}\left(P'_{\mathbf{c}_m}, \hat{P}_m\right) \\ &= \frac{1}{M}\sum_{m=1}^{M}\left(\frac{1}{\left|P'_{\mathbf{c}_m}\right|} \sum_{\mathbf{p}'_{ab}\in P'_{\mathbf{c}_m}} \min_{\hat{\mathbf{p}}_{ij}\in\hat{P}_m} \left\|\mathbf{p}'_{ab} - \hat{\mathbf{p}}_{ij}\right\|^2 \right. \\ &\qquad \left. + \frac{1}{\left|\hat{P}_m\right|} \sum_{\hat{\mathbf{p}}_{ij}\in\hat{P}_m} \min_{\mathbf{p}'_{ab}\in P'_{\mathbf{c}_m}} \left\|\mathbf{p}'_{ab} - \hat{\mathbf{p}}_{ij}\right\|^2 \right) \end{aligned} \tag{12}$$

where M represents the total number of vehicles. $P'_{\mathbf{c}_m}$ is a set of sampling positions of Cons-RIRConv, it represents the set of sampling positions when the center position is located at $\mathbf{c}_m$. $\mathbf{c}_m$ represents the center position of the mth oriented ground truth box, and L_{ch} represents the chamfer loss [40].

The Cons-RIRConv is depicted in Fig. 5. As depicted in Fig. 5, the parameters in the two transformation matrices, i.e., Δw, Δh, and θ, are obtained by applying a convolutional layer to the same input feature map. The regularization term L_{SRI} is used to constrain the set of transformed sampling positions. The Cons-RIRConv accomplishes the extraction of size- and rotation-invariant features of vehicles by transforming its sampling positions. Notably, Cons-RIRConv shares a similar concept with deformable convolution [36], which aims to adapt to geometric variations of targets by transforming the sampling positions of the convolutional kernel. However, Cons-RIRConv exhibits significant advantages in the task of SAR vehicle detection. First, Cons-RIRConv integrates the shape property of vehicles, adding rectangular constraints to the sampling positions, thereby reducing the difficulty of learning the sampling positions. Second, Cons-RIRConv requires learning only three parameters, i.e., Δw, Δh, and θ, during each convolution operation, whereas deformable convolution necessitates learning $K \times K \times 2$ parameters (where K is generally not less than 3). Therefore, Cons-RIRConv possesses fewer parameters and lower computational complexity compared to deformable convolution. Third, unlike deformable convolution, Cons-RIRConv is constrained by a clear supervisory signal, i.e., the size- and rotation-invariant regularization term, which further ensures the accuracy of the learned sampling positions.

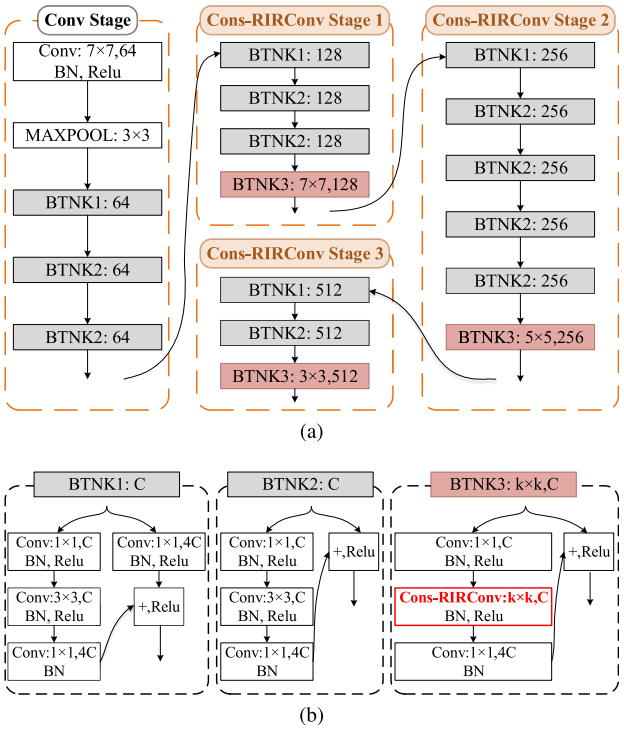

Fig. 6. Network structure of the backbone. (a) Overall structure of the backbone. (b) Specific structure of the bottleneck (BTNK).

We improve ResNet-50 [41] by incorporating Cons-RIRConv, thereby creating our backbone, as depicted in Fig. 6. Within the final bottleneck (BTNK) of each Cons-RIRConv stage, we employ Cons-RIRConv to enhance the ability of these stages to extract the size and rotation-invariant features of vehicles. Given the variable sizes of feature maps across different Cons-RIRConv stages, we adopt Cons-RIRConv with varying kernel sizes accordingly. The larger the input feature map, the larger the kernel size of Cons-RIRConv. This is crucial to ensure an adequate number of sampling positions for covering the vehicles.

C. Adaptive Anchor Generation Mechanism

To ensure high detection rates and reduce the regression complexity of OBB, many detection networks [18], [19], [20] densely sample a set of rotated anchor boxes with

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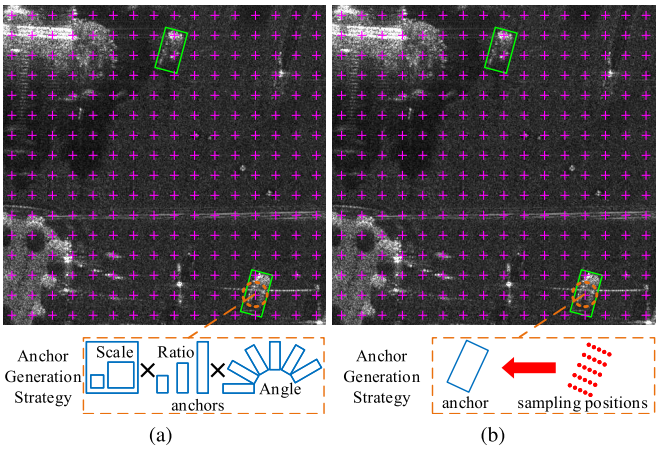

Fig. 7. Anchor generation strategies for both the traditional anchor-based method and the proposed method (taking the example of sampling anchor boxes at $16 \times 16 = 256$ positions in the input image). (a) Traditional anchor-based method. It sets anchor boxes with three scales, three aspect ratios, and six angles, resulting in a total of $16 \times 16 \times 3 \times 3 \times 6 = 13\,824$ anchor boxes on the input image. (b) Proposed method. It generates anchor boxes based on the sampling positions of Cons-RIRConv, with a total of $16 \times 16 \times 1 = 256$ anchor boxes on the input image.

fixed scales, aspect ratios, and angles in the input image, serving as "bridges" for OBB prediction, as depicted in Fig. 7(a). However, the excessive number of rotated anchor boxes poses a significant computational burden. In addition, given the sparse distribution of targets in SAR images, the majority of rotated anchor boxes serve as negative samples. Consequently, the issue of imbalanced positive and negative samples during training is pronouncedly evident, affecting the convergence of the network. Anchor-free methods [42], [43], [44] obviate the need for configuring rotated anchor boxes, thus alleviating the aforementioned issues. Nevertheless, due to the absence of rotated anchor boxes serving as "bridges," the detection performance of these methods tends to be unsatisfactory in SAR images with complex scenes. Therefore, in this article, we propose AAGM in the detection head of the network to simultaneously address all the aforementioned issues. A detailed explanation of AAGM is provided below.

From Section II-B, we know that through the application of scaling and rotation matrices, along with the imposition of constraints, the sampling positions of Cons-RIRConv closely align with the vehicles. This inspires us to devise a new anchor generation mechanism based on this characteristic of Cons-RIRConv. As depicted in Fig. 7(b), our newly designed AAGM eliminates the need for manual specification of scales, aspect ratios, and angles during the setting of rotated anchor boxes. Instead, it relies solely on the sampling positions of Cons-RIRConv for automatic generation. Specifically, each rotated anchor box generated by AAGM is represented based on the parameters in Cons-RIRConv, i.e., the center position $\mathbf{c}$ of sampling positions, the down-sampling factor s_r, the side length K of the convolution kernel, the scaling factors Δw and Δh, and the rotation angle θ

$$\mathrm{Ab}_c = \left\{ \left(c^x, c^y\right), \Delta w \times K/s_r, \Delta h \times K/s_r, \theta \right\} \tag{13}$$

where (c^x, c^y), $\Delta w \times K/s_r$, $\Delta h \times K/s_r$, θ denote the center coordinates, width, height, and angle of the rotated anchor box Ab_c, respectively. In AAGM, each pixel position of the $\mathbf{C}_{r\in[1,2,3]}$ serves as the center coordinate for generating rotated anchor boxes. During training, these rotated anchor boxes are assigned labels according to their intersection over union (IoU) with the ground truth boxes. Subsequently, the network computes the classification loss using (3) and the regression loss using (4), respectively.

Compared to traditional anchor generation strategies, AAGM offers two primary advantages. First, by eliminating the necessity to generate a vast number of rotated anchor boxes, which are typically based on predefined scales, aspect ratios, and angles, AAGM significantly reduces the redundancy of these boxes. This is advantageous for mitigating the computational burden and accelerating convergence. Second, rotated anchor boxes generated based on the sampling positions from Cons-RIRConv exhibit a higher degree of alignment with vehicles, thereby decreasing regression complexity and improving the localization accuracy of OBB.

D. Loss Function

In general, the overall loss function of the entire network comprises the classification loss L_{CLS}, the regression loss L_{REG}, and the size- and rotation-invariant regularization term L_{SRI}

$$L = L_{\mathrm{CLS}} + L_{\mathrm{REG}} + \lambda L_{\mathrm{SRI}} \tag{14}$$

where λ is the weight factor that balances the L_{CLS}, L_{REG}, and L_{SRI}, which is set to 0.5 by default.

III. Experiments

A. Dataset Description

We conducted experiments using the SIVED, miniSAR, and FARADSAR datasets, all of which are measured datasets specifically designed for SAR vehicle detection tasks. Detailed information regarding these three datasets is provided in Table I.

The SIVED dataset consists of 1044 SAR image chips sourced from Ku-, Ka-, and X-bands. Each chip has a size of 512×512, and each chip contains vehicle targets. Out of the total chips, 941 are used for training and validation, while the remaining 103 are served for testing. Unlike the SIVED dataset, the miniSAR dataset, and the FARADSAR dataset both consist of large-sized SAR images. Specifically, the miniSAR dataset comprises nine SAR images with the size of 1638×2510. Out of the total images, seven were chosen for training and validation, while the remaining two serve as test data. The FARADSAR dataset comprises 52 SAR images, with image sizes varying from 3842×1300 to 4082×1624. Out of the total images, 40 were chosen for training and validation, while the remaining 12 serve as test data. All images were manually annotated by SAR experts and have undergone rigorous, repeated verification by multiple individuals to guarantee the accuracy of the annotations. Figs. 8 and 9 exhibits examples from the three datasets.

Please note that the image sizes in the miniSAR and FARADSAR datasets are relatively large. The significant size of the original SAR image poses a considerable strain on

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TABLE I
CHARACTERISTICS OF THE SIVED, MINISAR, AND FARADSAR DATASETS

Parameter	SIVED	miniSAR	FARADSAR
Mode	Spotlight, Stripmap	Spotlight	Stripmap
Band	Ku, Ka, X	Ku	Ka
Range Resolution	4-inch, 0.3m	4-inch	4-inch
Azimuth Resolution	4-inch, 0.3m	4-inch	4-inch
Image size	512×512	1638×2510	3842×1300 to 4082×1624

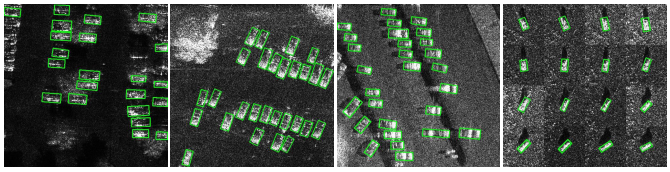
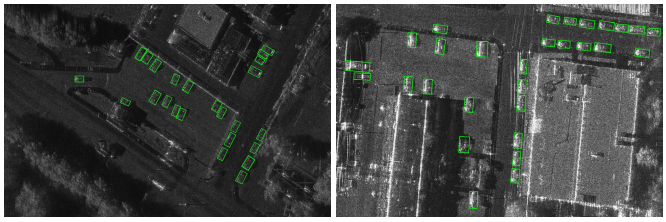


Fig. 8. Examples of SIVED dataset, where green OBBs represent the ground truth.



(a) (b)

Fig. 9. Examples of miniSAR dataset and FARADSAR dataset, where green OBBs represent the ground truth. (a) SAR image of miniSAR dataset. (b) SAR image of FARADSAR dataset.

graphics memory resources. Therefore, we employ a sliding window of 512×512 pixels with an overlap of 200 pixels to crop the large-sized images into multiple subimages. During testing, the detection results of the test subimages are stitched together to obtain the overall detection results for the original SAR image. It is noteworthy that, in the absence of any detected vehicle targets, the model will directly output an empty detection result. During training, subimages devoid of vehicle targets are discarded, and the remaining subimages are augmented by performing horizontal flipping, adding Gaussian noise, and applying rotations of 90°, 180°, and 270°. Finally, the miniSAR dataset comprises a total of 1463 subimages. The FARADSAR dataset comprises a total of 4454 subimages.

B. Experimental Settings

The MMRotate codebase [45] is utilized to implement the proposed method. The stochastic gradient descent algorithm with an initial learning rate of 0.001 is employed for optimizing the entire network. The network undergoes a total of 120 000 iterations during training. During testing, the non-maximum suppression algorithm [46] is employed to filter the overlapped OBB, with an IoU threshold of 0.3. The mini-batch size is 16 on two NVIDIA RTX 3090 GPUs.

C. Evaluation Metrics

In our experiments, we utilize four metrics, i.e., the false alarm rate (PFA), the detection rate (PD), the $F1$-score, and the mean average precision (mAP).

PFA and PD are defined as follows:

$$\text{PFA} = \frac{\text{FP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{PD} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

where FP represents the count of false alarms, TP represents the count of correctly detected vehicle targets, and FN corresponds to the count of missed detections. PFA quantifies the ratio of false positives to the total detected results. PD calculates the ratio of true positives to the total number of ground truths.

The $F1$-score is calculated from both the PFA and PD, providing a more comprehensive evaluation of a detection method

$$F1\text{-score} = \frac{2 \times (1 - \text{PFA}) \times \text{PD}}{(1 - \text{PFA}) + \text{PD}}. \tag{17}$$

The mAP aims to comprehensively reflect the quality of a detector, it models the area under precision–recall (PR) curves. Given the IoU of 0.5, mAP is defined as

$$\text{mAP} = \int_0^1 P(R) dR \tag{18}$$

where R and P represent recall and precision, respectively.

In these four metrics, a lower value of PFA, a higher value of PD, a higher value of the $F1$-score, and a higher value of the mAP indicate better performance of a detection method.

D. Comparison Between Different Detection Methods

We compare the proposed method with seven typical and powerful OBB-based methods. The comparison includes six anchor-based detectors: RRPN [18], OBB-based faster RCNN [15] (Faster RCNN-O), CSL [19], and R^3Det [20], KFIoU [47], and PSC [48], as well as three anchor-free detectors: OBB-based FCOS [42] (FCOS-O), OBB-based ATSS [43] (ATSS-O), and SASM RepPoints [44]. All the comparative methods employ ResNet-50 as their backbone. Based on the SIVED, miniSAR, and FARADSAR datasets, visualizations of the detection results for various methods are depicted in Figs. 10–12. The quantitative experimental results are presented in Tables II and III.

The ground scenes in the SAR images of SIVED, miniSAR, and FARADSAR datasets are highly complex, including numerous clutter such as trees, roadways, and buildings. As depicted in Fig. 10, the detection results obtained from the SIVED dataset are relatively good. All methods exhibit a modest number of false alarms and missed detections. The proposed method has the least number of false alarms and missed detections. As depicted in Figs. 11 and 12, this intricate clutter in large-sized SAR images poses significant challenges to all nine comparative methods. From Fig. 11(a), (b), and (g) as

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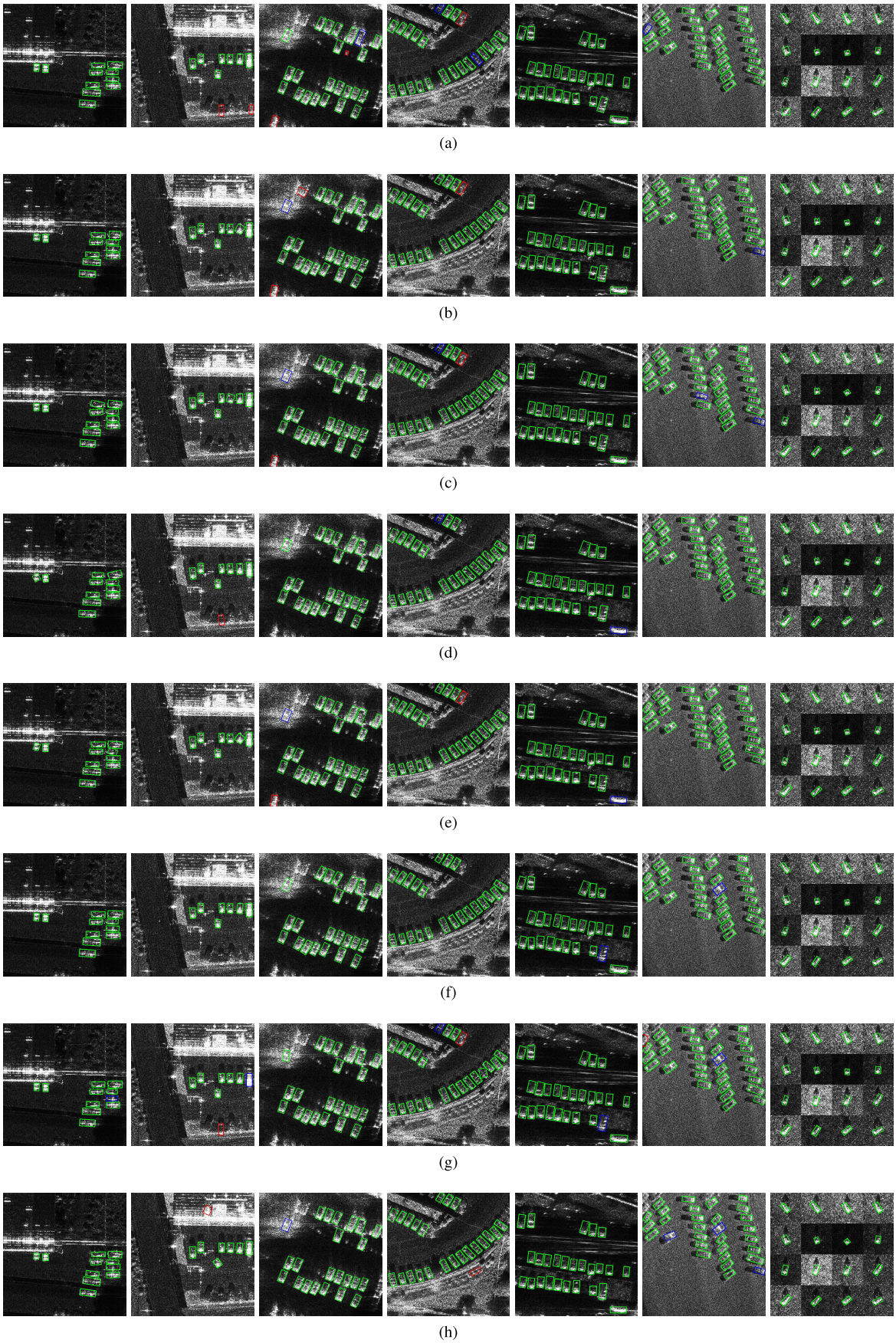

Fig. 10. Visualization comparison of various methods on seven images of SIVED, where the red, blue, and green OBBs, respectively, represent the false alarms, missed detections, and the correctly detected vehicle chips. (a) RRPN. (b) Faster RCNN-O. (c) CSL. (d) R3Det. (e) KFIoU. (f) PSC. (g) FCOS-O. (h) ATSS-O.

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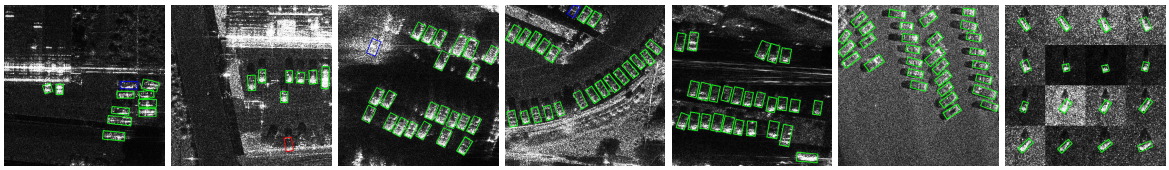
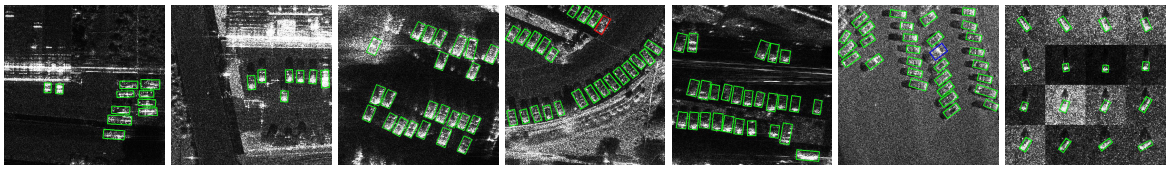


(i)



(j)

Fig. 10. *(Continued.)* Visualization comparison of various methods on seven images of SIVED, where the red, blue, and green OBBs, respectively, represent the false alarms, missed detections, and the correctly detected vehicle chips. (i) SASM RepPoints. (j) Proposed method.

TABLE II
QUANTITATIVE EXPERIMENTAL RESULTS OF VARIOUS METHODS ON SIVED DATASET

Method	PFA	PD	F1-score	mAP	Params (M)	FLOPs (G)
RRPN	0.0951	0.9512	0.9275	0.9536	41.14	63.67
Faster RCNN-O	0.0494	0.9691	0.9597	0.9798	41.12	63.25
CSL	0.0482	0.9634	0.9576	0.9726	37.08	57.61
R^3Det	0.0474	0.9797	0.9659	0.9886	41.91	83.93
KFIoU	0.0513	0.9756	0.9619	0.9833	41.61	82.31
PSC	0.0436	0.9813	0.9687	0.9911	36.27	53.19
FCOS-O	0.0500	0.9260	0.9378	0.9637	31.89	51.56
ATSS-O	0.0496	0.9667	0.9585	0.9755	36.02	51.81
SASM RepPoints	0.0420	0.9634	0.9607	0.9817	36.61	**48.56**
Proposed method	**0.0351**	**0.9821**	**0.9734**	**0.9925**	**30.88**	51.93

TABLE III
QUANTITATIVE EXPERIMENTAL RESULTS OF VARIOUS METHODS ON MINISAR AND FARADSAR DATASETS

Method	miniSAR dataset				FARADSAR dataset				Params	FLOPs
	PFA	PD	F1-score	mAP	PFA	PD	F1-score	mAP	(M)	(G)
RRPN	0.1880	0.8871	0.8479	0.8166	0.1732	0.8068	0.8167	0.7628	41.14	63.67
Faster RCNN-O	0.1706	0.8629	0.8458	0.8259	0.2193	**0.8939**	0.8335	0.7861	41.12	63.25
CSL	0.1497	0.8790	0.8645	0.8356	0.1750	0.8428	0.8338	0.7889	37.08	57.61
R^3Det	0.1334	0.8629	0.8647	0.8402	0.1613	0.8371	0.8379	0.7986	41.91	83.93
KFIoU	0.1323	0.8629	0.8653	0.8332	0.1692	0.8314	0.8311	0.7931	41.61	82.31
PSC	0.1407	0.8952	0.8769	0.8508	0.1321	0.8428	0.8552	0.8004	36.27	53.19
FCOS-O	0.1429	0.8226	0.8395	0.8037	0.1928	0.8447	0.8256	0.7757	31.89	51.56
ATSS-O	0.1312	0.8548	0.8617	0.8321	0.1119	0.7992	0.8414	0.7935	36.02	51.81
SASM RepPoints	0.1520	0.8548	0.8514	0.8115	0.1342	0.8106	0.8373	0.7842	36.61	**48.56**
Proposed method	**0.1016**	**0.9274**	**0.9127**	**0.8759**	**0.0915**	0.8883	**0.8982**	**0.8408**	**30.88**	51.93

well as Fig. 12(a), (b), and (g), it is evident that RRPN, FCOS-O, and Faster RCNN-O exhibit a high number of missed detections and false alarms. The detection results of these three methods are the poorest among the evaluated methods.

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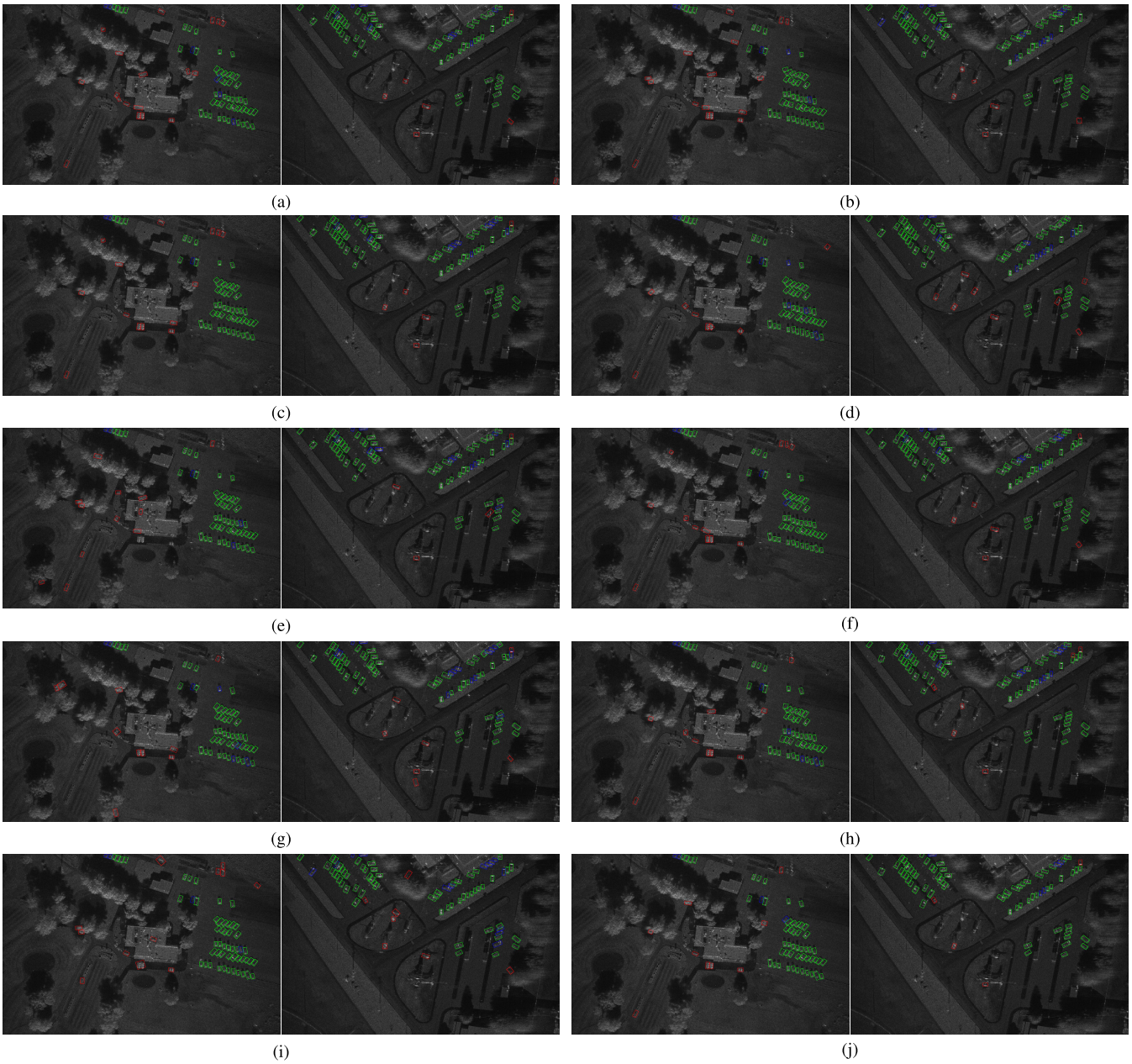

Fig. 11. Visualization comparison of various methods on two images of miniSAR, where the red, blue, and green OBBs, respectively, represent the false alarms, missed detections, and the correctly detected vehicle chips. (a) RRPN. (b) Faster RCNN-O. (c) CSL. (d) R3Det. (e) KFIoU. (f) PSC. (g) FCOS-O. (h) ATSS-O. (i) SASM RepPoints. (j) Proposed method.

In comparison, the CSL, R^3Det, KFIoU, PSC, ATSS-O, and SASM RepPoints demonstrate relatively superior performance. The R^3Det, PSC, and ATSS-O, in particular, achieve relatively better detection results on both the miniSAR and FARADSAR datasets compared to the other methods. Nevertheless, their detection results still exhibit a notable number of missed detections and false alarms. As evident from Figs. 11(j) and 12(j), the proposed method outperforms all comparative methods. It exhibits a small number of missed detections and demonstrates excellent detection performance for vehicles of various sizes and angles. Moreover, it generates the least number of false alarms in complex ground scenes compared to other methods.

As evident from Table II, all methods achieve excellent detection performance on the SIVED dataset, with $F1$-scores and mAPs exceeding 90%. In comparison to other methods, the proposed method has the lowest PFA and the highest PD, $F1$-score, and mAP. From Table III, we can observe that our proposed method achieves the lowest PFA across both miniSAR and FARADSAR datasets compared to the nine comparative methods. Moreover, our proposed method attains the highest PD on the miniSAR dataset and ranks second only to Faster RCNN-O on the FARADSAR dataset. Upon evaluating the more comprehensive $F1$-score and mAP metrics, our proposed method outperforms all others, attaining the top $F1$-score and the top mAP on miniSAR and FARADSAR

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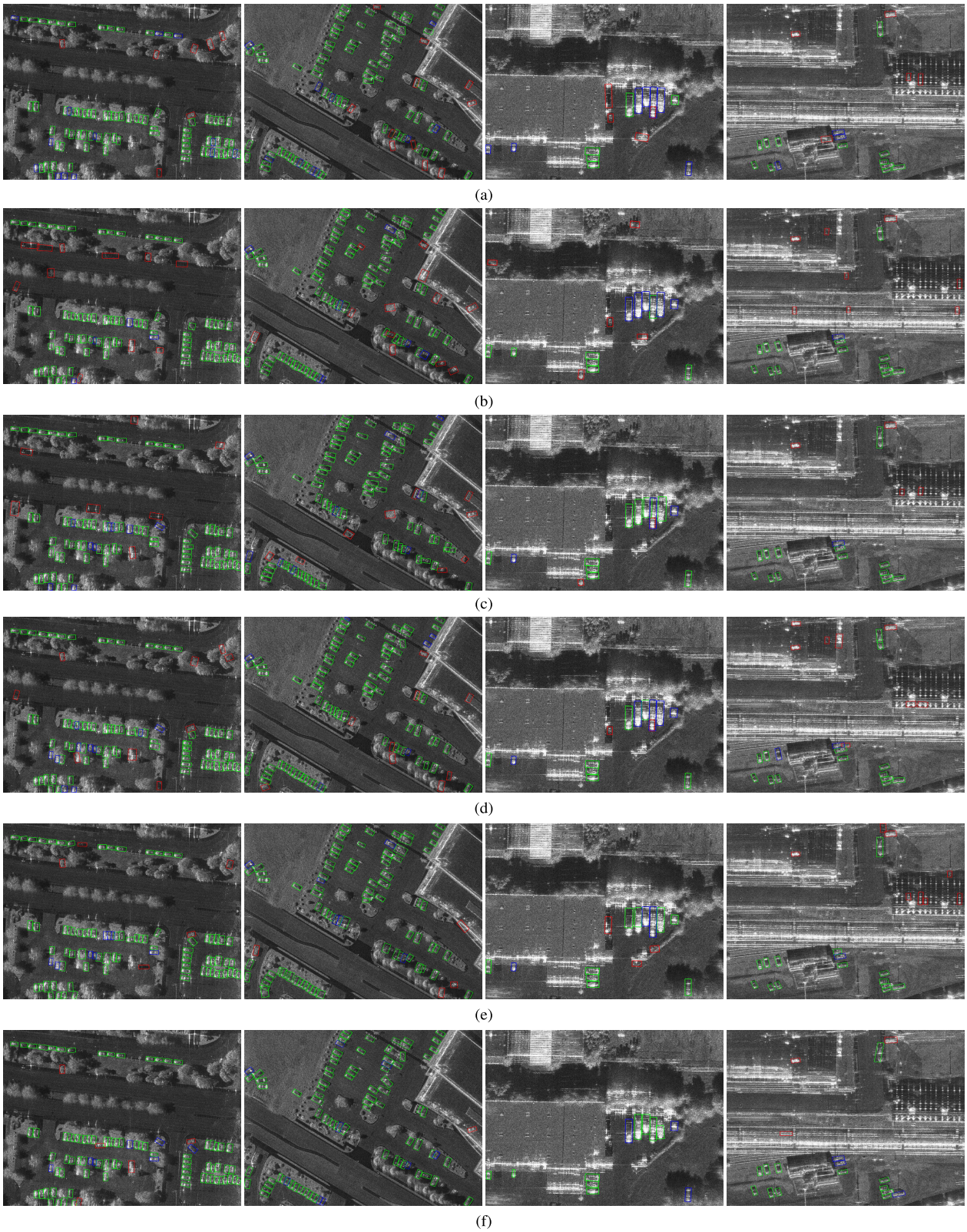

Fig. 12. Visualization comparison of various methods on four images of FARADSAR, where the red, blue, and green OBBs, respectively, represent the false alarms, missed detections, and the correctly detected vehicle chips. (a) RRPN. (b) Faster RCNN-O. (c) CSL. (d) R3Det. (e) KFIoU. (f) PSC.

datasets. In addition, among all the methods, the proposed method has the lowest number of parameters (Params), and its floating point operations (FLOPs) are slightly lower than those of FCOS-O, ATSS-O, and SASM RepPoints. The reasons are as follows: by combining the size property of vehicles and the ITS of the feature maps, the proposed method necessitates retaining only three distinct scales of feature maps within the feature pyramid. However, most other methods employ four

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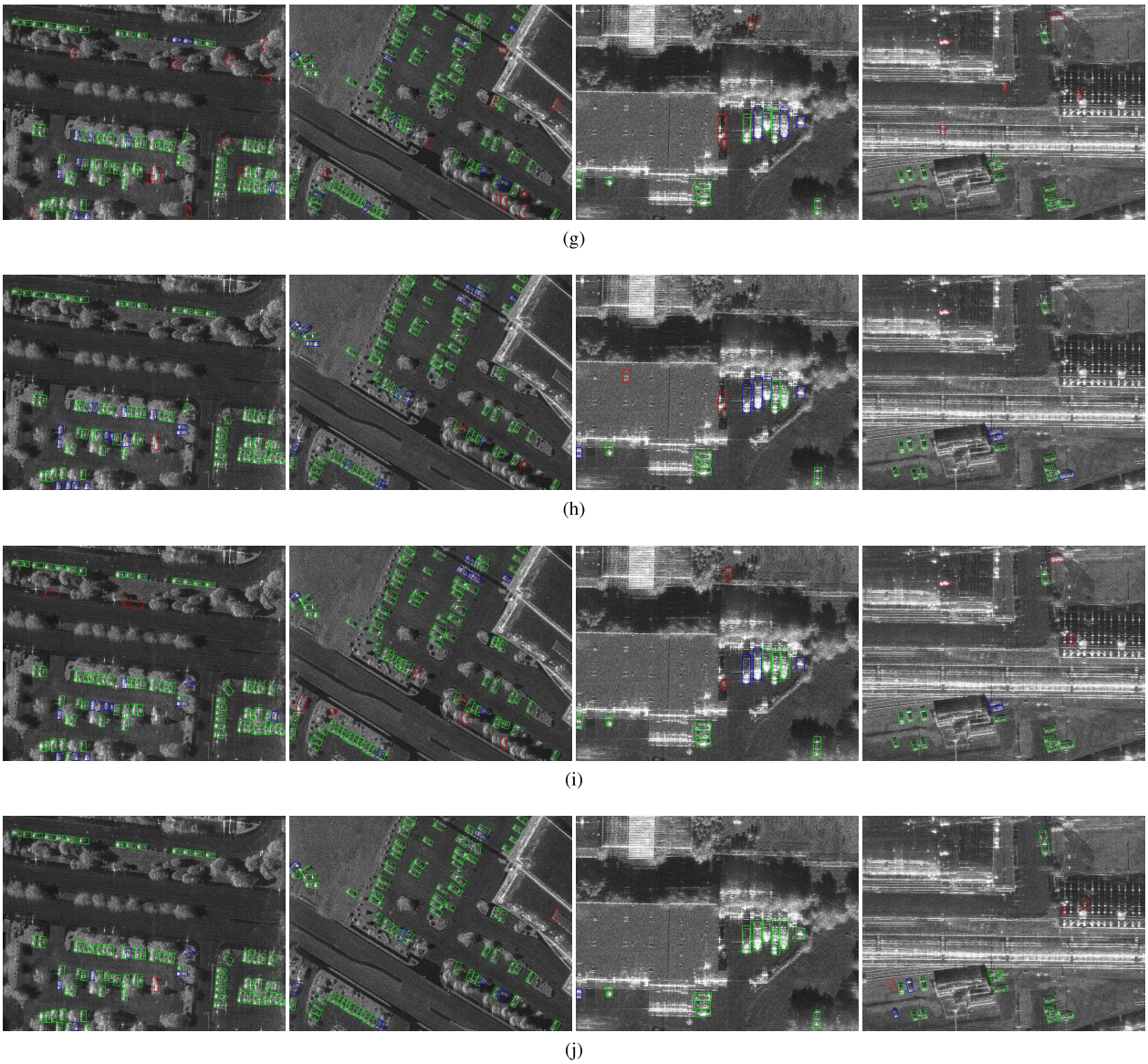

Fig. 12. *(Continued.)* Visualization comparison of various methods on four images of FARADSAR, where the red, blue, and green OBBs, respectively, represent the false alarms, missed detections, and the correctly detected vehicle chips. (g) FCOS-O. (h) ATSS-O. (i) SASM RepPoints. (j) Proposed method.

or five different scales of feature maps, resulting in unnecessary parameters and computational redundancy for vehicle detection. Consequently, it can be concluded that, compared to other methods, the proposed method demonstrates superior performance.

E. Model Analysis

In this section, we conduct a series of experiments on the FARADSAR dataset to validate the effectiveness of both Cons-RIRConv and AAGM.

1) Effect of Cons-RIRConv: Cons-RIRConv is designed based on the shape property of vehicles, comprising two key components: the transformation of sampling positions and the size- and rotation-invariant regularization term. We conduct ablation experiments to analyze the role of each component. In addition, in order to verify the advantages of Cons-RIRConv over other convolution operations, we substitute the ConsRIRConv utilized in the proposed network with standard convolution and deformable convolution [36], and conduct experimental comparisons. Deformable convolution models the geometric transformations of targets by learning offsets for each sampling position, yet it does not incorporate the shape property of targets. Table IV presents the quantitative experimental results obtained using different convolution operations. It is noteworthy that, to guarantee a fair comparison in experiments, all methods listed in Table IV adopt the same conventional anchor generation strategy. Precisely, the scales, aspect ratios, and angles of the anchors are manually predefined rather than automatically generated using AAGM. To visually illustrate the variation in sampling positions during the extraction of vehicle features, we visualized the sampling positions of the 7×7 Cons-RIRConv kernel as it slides into the center of the vehicle. In addition, we also visualized the sampling positions after replacing Cons-RIRConv with standard convolution and deformable convolution, as shown in Fig. 13.

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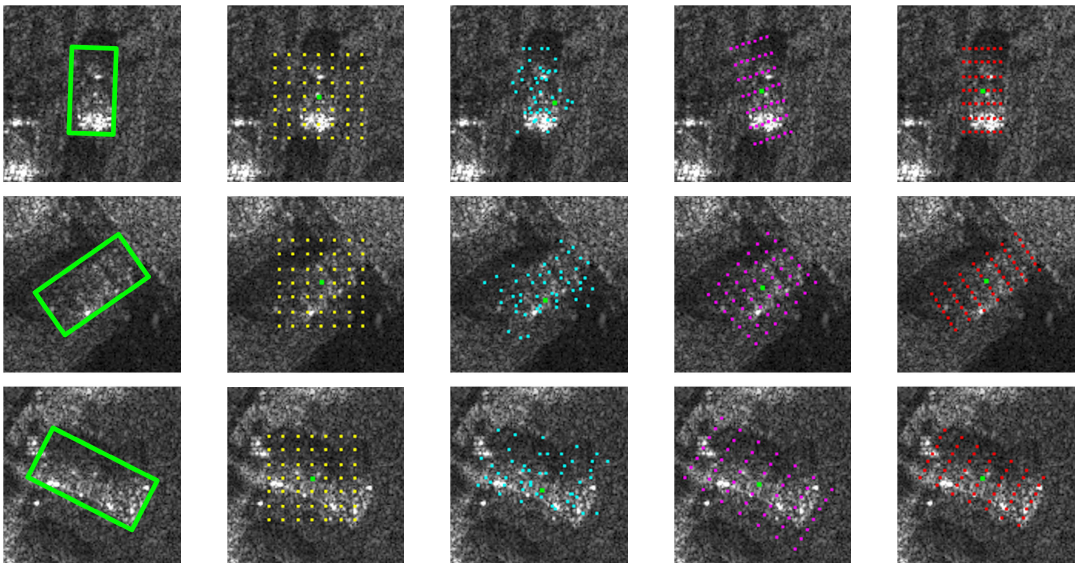
TABLE IV
QUANTITATIVE EXPERIMENTAL RESULTS USING DIFFERENT CONVOLUTION OPERATIONS

	Transform Sampling Positions	Shape property	Regularization Term	PFA	PD	F1-score	Params (M)	FLOPs (G)
Standard Conv	×	×	×	0.1568	0.8352	0.8392	**31.71**	**56.50**
Deform Conv	√	×	×	0.1670	0.8788	0.8553	32.02	57.11
Cons-RIRConv w/o regularization term	√	√	×	0.1359	0.8693	0.8667	31.74	56.53
Cons-RIRConv	√	√	√	**0.1293**	**0.8844**	**0.8776**	31.74	56.53



(a) (b) (c) (d) (e)

Fig. 13. Visualization of sampling positions in different convolution operations, where the center of the sampling positions is indicated by green dots. (a) Ground truth of SAR vehicles. (b) Sampling positions in the standard convolution. (c) Sampling positions in the deformable convolution. (d) Sampling positions in the Cons-RIRConv w/o regularization term. (e) Sampling positions in the Cons-RIRConv.

TABLE V
QUANTITATIVE EXPERIMENTAL RESULTS WHEN SETTING DIFFERENT KERNEL SIZES IN CONS-RIRCONV

(K_1, K_2, K_3)	PFA	PD	F1-score
$(7, 5, 5)$	0.1286	0.8769	0.8742
$(7, 5, 7)$	0.1424	0.8731	0.8653
$(7, 3, 3)$	0.1466	0.8769	0.8650
$(7, 7, 3)$	0.1246	0.8693	0.8724
$(3, 5, 3)$	0.1523	0.8769	0.8621
$(5, 5, 3)$	**0.1219**	0.8655	0.8718
$(7, 5, 3)$	0.1293	**0.8844**	**0.8776**

As evident from Table IV, the utilization of the standard convolution operation yields the poorest detection performance in comparison to other methods. By introducing offsets for each original sampling position, deformable convolution enhances its robustness to geometric transformations of vehicles, thus improving detection performance to some extent. However, due to the absence of any prior information, the sampling positions of deformable convolution exhibit irregular patterns and still fail to align well with vehicles, as shown in Fig. 13(c). The Cons-RIRConv w/o regularization term and Cons-RIRConv, by incorporating the shape property of vehicles, exhibit sampling positions that maintain a rectangular shape while adapting to variations in vehicle size and orientation through scaling and rotation transformations, as illustrated in Fig. 13(d) and (e). Therefore, they achieve superior detection performance compared to deformable convolution, as indicated in Table IV, rows 4 and 5. By applying the size- and rotation-invariant regularization term, the scaling and rotation matrices in Cons-RIRConv are learned with greater precision, thereby leading to sampling positions that are more closely aligned with the vehicles. In terms of the Params and FLOPs, it can be observed that the Params and FLOPs of Cons-RIRConv are slightly higher than those of Standard Conv but remain lower than those of Deform Conv. This signifies that Cons-RIRConv achieves significant performance improvement with only a minimal increase in Params and FLOPs. Therefore, compared to other convolution operations, Cons-RIRConv achieves the most outstanding detection performance. These results validate the effectiveness of incorporating the shape property of vehicles, transforming the sampling positions, and

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TABLE VI
QUANTITATIVE EXPERIMENTAL RESULTS WITH DIFFERENT ANCHOR GENERATION STRATEGIES

	PFA	PD	F1-score	Params (M)	FLOPs (G)	Test time (seconds)
Traditional anchor generation strategy	0.1293	0.8844	0.8776	31.74	56.53	107.5
AAGM	**0.0915**	**0.8883**	**0.8982**	**30.88**	**51.93**	**98.6**

designing the size- and rotation-invariant regularization term in Cons-RIRConv.

2) Influence of the Kernel Size in Cons-RIRConv: In Section II-B, given the varying sizes of feature maps across different Cons-RIRConv stages, we set the kernel sizes for Cons-RIRConv in Cons-RIRConv Stage 1, Stage 2, and Stage 3 to 7×7, 5×5, and 3×3, respectively. In this section, we vary the kernel sizes in Cons-RIRConv to investigate their influence on detection performance. Table V presents the corresponding experimental results.

In Table V, (K_1, K_2, K_3) indicates that the kernel sizes for Cons-RIRConv in Cons-RIRConv Stage 1, Stage 2, and Stage 3 are set to $K_1 \times K_1$, $K_2 \times K_2$, and $K_3 \times K_3$, respectively. From the second, third, and eighth rows of Table V, we can observe that setting the kernel size of Cons-RIRConv in Cons-RIRConv Stage 3 to 3×3 achieves higher detection performance. Similarly, conclusions can be drawn from the results in the fourth, fifth, and eighth rows, as well as the sixth, seventh, and eighth rows of Table V. These indicate that the most effective kernel sizes for Cons-RIRConv in Cons-RIRConv Stage 1 and Stage 2 are 7×7 and 5×5, respectively. We believe the reasons for this may be as follows: for larger input feature maps, where vehicles occupy a greater number of feature points, it is necessary to set relatively larger kernels to ensure extensive coverage of sampling positions across the vehicle area. Conversely, for smaller input feature maps, where vehicles occupy fewer feature points, utilizing smaller kernels is advantageous as it minimizes interference from surrounding cluttered regions.

3) Effect of AAGM: The AAGM integrates the anchor box generation process of the model with the sampling positions of Cons-RIRConv, thereby enabling it to adaptively generate rotation anchor boxes that exhibit low redundancy and a high degree of alignment with vehicles. To assess the improvements in both detection performance and speed achieved through the implementation of AAGM, we have conducted experiments, and the corresponding results are presented in Table VI. Notably, the traditional anchor generation strategy remains consistent with [18], with the scales, aspect ratios, and angles of the anchors being manually predefined. Specifically, we set three different scales (8, 16, and 32), three different aspect ratios (1:2, 1:5, and 1:8), and six different angles ($-30°$, $0°$, $30°$, $60°$, $90°$, and $120°$). Consequently, for the traditional anchor generation strategy, 54 anchor boxes are generated for every point on the feature map.

Table VI compares the detection performance and total test time on all test images of FARADSAR between the two methods. It is evident that, in comparison to the traditional anchor box generation strategy, the employment of AAGM leads to a reduction in the PFA and an improvement in the PD, thereby enhancing the overall detection performance. Furthermore, the significant reduction in anchor redundancy achieved by AAGM enhances the computational efficiency of the model, subsequently reducing the Params, FLOPs, and total test time. Fig. 14 shows the comparison results between the traditional anchor generation strategy and the AAGM. Evidently, compared to the traditional anchor generation strategy, the predicted OBBs through AAGM exhibit a higher alignment

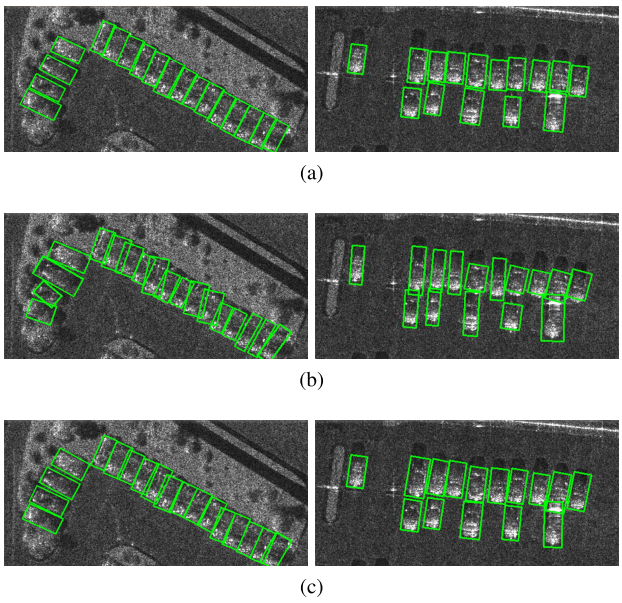

Fig. 14. Comparison results of the traditional anchor generation strategy and the AAGM. (a) Ground truth of SAR vehicles. (b) Detection results of the traditional anchor generation strategy. (c) Detection results of the AAGM.

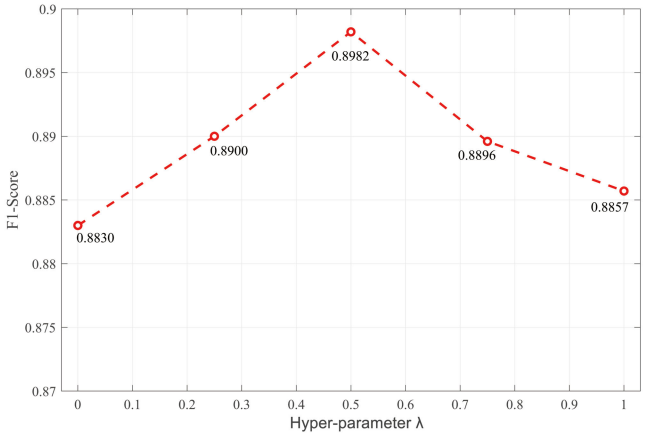

Fig. 15. Influence of the weight factor λ on detection performance.

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with the vehicles. This indicates that AAGM can enhance the localization accuracy of the predicted OBB.

4) Influence of the Weight Factor λ in Overall Loss Function: To assess the influence of the weighting factor λ on detection performance, we plotted $F1$-score curves for different values of λ, as depicted in Fig. 15. The results indicate that the model attains the best detection performance when λ is adjusted to 0.5. The reasons might be as follows: when λ is relatively small, the influence of the size- and rotation-invariant regularization term is muted, resulting in insufficient learning of sampling positions by Cons-RIRConv. Consequently, it is challenging to achieve high detection performance. Conversely, when λ is relatively large, the focus of the model on learning for classification and regression tasks is somewhat diminished, also affecting detection performance.

IV. Conclusion

We propose a novel-oriented SAR vehicle detection network in this article. It incorporates the size and shape properties of vehicles. Within the backbone of the network, we introduce Cons-RIRConv, which incorporates the shape property of vehicles. By adaptively adjusting its convolution sampling positions, Cons-RIRConv is able to extract size- and rotation-invariant features of vehicles, thereby improving detection performance. Furthermore, the proposed method integrates the sampling positions of Cons-RIRConv into the generation process of rotation anchor boxes, thus facilitating the design of the AAGM. Guided by the sampling positions of Cons-RIRConv, AAGM improves the localization accuracy of OBB and reduces computational costs. Experiments on three authoritative measured SAR vehicle detection datasets show the effectiveness of our method.

References

[1] L. M. Novak, M. C. Burl, and W. W. Irving, "Optimal polarimetric processing for enhanced target detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 29, no. 1, pp. 234–244, Jan. 1993.

[2] G. Gao, L. Liu, L. Zhao, G. Shi, and G. Kuang, "An adaptive and fast CFAR algorithm based on automatic censoring for target detection in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 6, pp. 1685–1697, Jun. 2009.

[3] T. Zhang, X. Zhang, J. Shi, and S. Wei, "HyperLi-Net: A hyper-light deep learning network for high-accurate and high-speed ship detection from synthetic aperture radar imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 123–153, Sep. 2020.

[4] Z. Cui, Q. Li, Z. Cao, and N. Liu, "Dense attention pyramid networks for multi-scale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8983–8997, Nov. 2019.

[5] X. Sun, Y. Lv, Z. Wang, and K. Fu, "SCAN: Scattering characteristics analysis network for few-shot aircraft classification in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5226517.

[6] Q. Guo, H. Wang, and F. Xu, "Scattering enhanced attention pyramid network for aircraft detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7570–7587, Sep. 2021.

[7] C. Liu, Y. Wang, and J. Guo, "Bridge detection in polarimetric SAR images based on water network construction," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Kuala Lumpur, Malaysia, Jul. 2022, pp. 7879–7882.

[8] C. Marin, F. Bovolo, and L. Bruzzone, "Building change detection in multitemporal very high resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2664–2682, May 2015.

[9] L. Du, L. Li, D. Wei, and J. Mao, "Saliency-guided single shot multibox detector for target detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3366–3376, May 2020.

[10] Z. Wang, L. Du, and H. Su, "Target detection via Bayesian-morphological saliency in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5455–5466, Oct. 2017.

[11] L. Zhai, Y. Li, and Y. Su, "Inshore ship detection via saliency and context information in high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1870–1874, Dec. 2016.

[12] M. Tello, C. Lopez-Martinez, and J. J. Mallorqui, "A novel algorithm for ship detection in SAR imagery based on the wavelet transform," *IEEE Geosci. Remote Sens. Lett.*, vol. 2, no. 2, pp. 201–205, Apr. 2005.

[13] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vis.*, Oct. 2016, pp. 21–37.

[14] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.

[15] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Int. Conf. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[16] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[17] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[18] J. Ma et al., "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.

[19] X. Yang and J. Yan, "Arbitrary-oriented object detection with circular smooth label," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 677–694.

[20] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 4, pp. 3163–3171.

[21] T. Zhang et al., "Balance learning for ship detection from synthetic aperture radar remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 182, pp. 190–207, Dec. 2021.

[22] Y. Du, L. Du, Y. Guo, and Y. Shi, "Semisupervised SAR ship detection network via scene characteristic learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5201517.

[23] F. Gao, Y. He, J. Wang, A. Hussain, and H. Zhou, "Anchor-free convolutional network with dense attention feature aggregation for ship detection in SAR images," *Remote Sens.*, vol. 12, no. 16, p. 2619, Aug. 2020.

[24] X. Ma, S. Hou, Y. Wang, J. Wang, and H. Wang, "Multiscale and dense ship detection in SAR images based on key-point estimation and attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5221111.

[25] X. Zheng, H. Cui, C. Xu, and X. Lu, "Dual teacher: A semisupervised cotraining framework for cross-domain ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613312.

[26] L. Liu, Z. Pan, and B. Lei, "Learning a rotation invariant detector with rotatable bounding box," 2017, *arXiv:1711.09405*.

[27] Q. An, Z. Pan, L. Liu, and H. You, "DRBox-V2: An improved detector with rotatable boxes for target detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8333–8349, Nov. 2019.

[28] K. Fu, J. Fu, Z. Wang, and X. Sun, "Scattering-keypoint-guided network for oriented ship detection in high-resolution and large-scale SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11162–11178, 2021.

[29] F. Gao, Y. Huo, J. Sun, T. Yu, A. Hussain, and H. Zhou, "Ellipse encoding for arbitrary-oriented SAR ship detection based on dynamic key points," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5240528.

[30] J. Zhang, M. Xing, G.-C. Sun, and N. Li, "Oriented Gaussian function-based box boundary-aware vectors for oriented ship detection in multiresolution SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5211015.

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[31] Z. Sun, X. Leng, Y. Lei, B. Xiong, K. Ji, and G. Kuang, "BiFA-YOLO: A novel YOLO-based method for arbitrary-oriented ship detection in high-resolution SAR images," *Remote Sens.*, vol. 13, no. 21, p. 4209, Oct. 2021.

[32] L. Chen et al., "Method and system for bridge detection in SAR images using rotated bounding box-based dynamic global attention," Patent CN 2 022 111 304 698, Mar. 7, 2023.

[33] *SANDIA Mini SAR Complex Imagery*. Accessed: 2006. [Online]. Available: https://www.sandia.gov/radar/complex-data/index.html

[34] *FARADSAR Public Release Data*. Accessed: 2015. [Online]. Available: https://www.sandia.gov/radar/complex-data/FARAD-KA-BAND.zip

[35] X. Lin, B. Zhang, F. Wu, C. Wang, Y. Yang, and H. Chen, "SIVED: A SAR image dataset for vehicle detection based on rotatable bounding box," *Remote Sens.*, vol. 15, no. 11, p. 2825, May 2023.

[36] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 764–773.

[37] L. Li, Y. Du, and L. Du, "Vehicle target detection network in SAR images based on rectangle-invariant rotatable convolution," *Remote Sens.*, vol. 14, no. 13, p. 3086, Jun. 2022.

[38] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[39] R. Yang, Z. Pan, X. Jia, L. Zhang, and Y. Deng, "A novel CNN-based detector for ship detection based on rotatable bounding box in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1938–1958, 2021.

[40] H. Fan, H. Su, and L. Guibas, "A point set generation network for 3D object reconstruction from a single image," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2463–2471.

[41] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[42] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Seoul, South Korea, Oct. 2019, pp. 9626–9635.

[43] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 9756–9765.

[44] L. Hou, K. Lu, J. Xue, and Y. Li, "Shape-adaptive selection and measurement for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, vol. 36, no. 1, pp. 923–932.

[45] Y. Zhou et al., "MMRotate: A rotated object detection benchmark using PyTorch," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 7331–7334.

[46] A. Neubeck and L. Van Gool, "Efficient non-maximum suppression," in *Proc. 18th Int. Conf. Pattern Recognit. (ICPR)*, Jun. 2006, pp. 850–855.

[47] X. Yang et al., "The KFIoU loss for rotated object detection," in *Proc. 11th Int. Conf. Learn. Represent. (ICLR)*, Jan. 2022, pp. 1–9.

[48] Y. Yu and F. Da, "Phase-shifting coder: Predicting accurate orientation in oriented object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, Jun. 2023, pp. 13354–13363.

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